Applications of ICT

LAB REPORT # 02

Submitted by

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Introduction

For loops in python are used to repeat a set of code a specific number of times. Range function is used to implement for loop.

Objectives

The purpose of this lab was to understand the concept of repetition strcutures like for loop and its implementation using the range() function in python.

- **Understanding for loop**
- **Understanding range() function**

For loop and range():

For loop is used to repeat a set of code specific number of times in such a way that a counter variable is used that is incremented every time until it reaches the limit number that we specify. To implement a for loop in python, we use range() function. range(start, stop, step). range usually takes 3 integers as parameter tho it can use 1 or 2 asw. Start is the variable that contains the integer that starts the loop, it is incremented by step every time the loop runs. The loop stops once the start variable is incremented enough i.e start becomes equal to stop.

Example:

Suppose we have to print a string 500 times. Instead of writing print(string) 500 times we can use a repetition structure such as for loop.

Syntax for a for loop is as follows:

```
for x in range(start, stop, step):
    Code
```

So, in this example:

```
for i in range(0,500,1):
    print("hello world")
```

Tasks

Task 1:

Write a program in python that gets a string from user and checks whether the user entered data is a ***palindrome or not.*** E.g.,

- User enters: LEVEL, It is a palindrome as Reverse of this remains LEVEL
- User enters: 12344321, It is a palindrome as Reverse of this remains 12344321

The code such that it re-runs until user enters No or Quit

CODE:

```
Continue = True
while Continue:
    print("Enter a word (Enter no or quit to close the program)")
    userInput = input("Enter ur choice: ").lower()
    if userInput == "no" or userInput == "quit":
        break
    else:
        Palindrome = True
        length = len(userInput)
        Index = len(userInput)-1
        for i in range(0, length):
            if userInput[i] != userInput[Index]:
                Palindrome = False
            if i+1==Index:
                break
            Index -=1
        if Palindrome:
            print("Pallindrome")
        else:
            print("Not pallindrome")
```

OUTPUT:

```
Enter a word (Enter no or quit to close the program)
Enter ur choice: level
Pallindrome
Enter a word (Enter no or quit to close the program)
Enter ur choice: hello
Not pallindrome
Enter a word (Enter no or quit to close the program)
Enter ur choice: no
```

Task 2:

We want to create a program that can automatically calculate the score in Scrabble. Write a program that takes a string of letters and calculates the total word score using a if/else statements.

Note: We are using a subset of the English alphabet.

Letter	Point Value
Q,Z	10
J,X	8
K	5
F,H,V	4
B,C,M,P	3
D, G	2
A, E, I, O, U	1

CODE:

```
userInput = input("Enter a word: ").lower()
length = len(userInput)
sum = 0
for i in range(length):
    if userInput[i] == "q" or userInput[i] == "z":
        sum = sum + 10
    elif userInput[i] == "j" or userInput[i] == "x":
        sum = sum + 8
    elif userInput[i] == "k":
        sum = sum + 5
    elif userInput[i] == "f" or userInput[i] == "h" or userInput[i] == "v":
        sum = sum + 4
    elif userInput[i] == "b" or userInput[i] == "c" or userInput[i] == "m" or userInput == "p":
        sum = sum + 3
    elif userInput[i] == "d" or userInput[i] == "g":
        sum = sum + 2
```

```
    elif userInput[i] == "a" or userInput[i] == "e" or userInput[i] == "i" or userInput[i] == "o" or
userInput[i] == "u":
        sum = sum + 1
print(sum)
```

OUTPUT:

```
Enter a word: Unsab
5
```

Task 3:

You are required to create a python program for calculating Combinations and Permutations using the following formulae:

$$\text{Combinations: } \binom{n}{r} = \frac{n!}{r! \cdot (n-r)!}$$

$$\text{Permutations: } P_r = \frac{n!}{(n-r)!}$$

The factorial in the above is calculated as:

$$n! = \prod_{i=1}^{n} i$$

For example: 5! = 1*2*3*4*5 = 120

CODE:

```
n = int(input("Enter n : "))
r = int(input("Enter r : "))
nFactorial = 1
rFactorial = 1
subFactorial = 1
if r < 0:
    print("Invalid input")
elif r > n:
    print("Combinations : 0")
    print("Permutations : 0")
else:
    for i in range(n):
        nFactorial = nFactorial * (1+i)
    for x in range(r):
        rFactorial = rFactorial * (1+x)
    for y in range(n-r):
        subFactorial = subFactorial * (1+y)
```

```
    Combinations = nFactorial / (rFactorial * subFactorial)
    Permutations = nFactorial / subFactorial

    print(Combinations)
    print(Permutations)
```

OUTPUT:

```
Enter n : 10
Enter r : 4
210.0
5040.0
```

Observations

I observed in this lab that python offers for loop for efficient code such that it saves time as well as the lines of code written that also reduces the memory used in storing a program.

Conclusion

This lab was useful as it taught us about repetition structures and their implementation in python. Specifically, for loop.